Bureau
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The International System of Units

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Annexe 2
Annex 2

Valeurs recommandées des fréquences étalons destinées à la mise en pratique de la définition du mètre et aux représentations secondaires de la seconde

Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second

Partie 2
Part 2

Rubidium ()

Rubidium ()

CCL-CCTF Frequency Standards Working Group

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Part 2: Rubidium ()

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Appendix 2

1. CIPM recommended values

The values $f = 385\ 285\ 142\ 375\ \mathrm{kHz}$

$\lambda = 778\ 105\ 421.23\ \mathrm{fm}$

with a relative standard uncertainty of 1.3×10^{-11} apply to the radiation of a laser stabilized to the centre of the twophoton transition. The values apply to a rubidium cell at a temperature below 100 °C and are corrected to zero laser power.

2. Source data

Adopted value

$f = 385\ 285\ 142\ 375\ (5)$ kHz

$u_c/y = 1.3 \times 10^{-11}$

for which:

$\lambda = 778\ 105\ 421.23\ (1)$ fm

$u_c/y = 1.3 \times 10^{-11}$

calculated from

f/ kHz	u_c/y	source data
385 285 142 378.3	5.2×10^{-12}	[1]
385 285 142 373.8	3.4×10^{-12}	[2]
385 285 142 372.3	1.4×10^{-11}	Section 2.1
Weighted mean	$f = 385\ 285\ 142\ 375.0$ kHz	

applies to the single-photon laser frequency of the two-photon transition. The CCL decided to attribute a standard uncertainty of 5 kHz, corresponding to a relative standard uncertainty of 1.3×10^{-11}.(1)

(1) A recent measurement made after the CCL 2001 has confirmed one of the data [3].

2.1.

Bernard et al. [4] gives

$$f\{^{87}\mathrm{Rb},\ 5\mathrm{S}_{1/2}\ (\mathrm{F_g}=2) - 5\mathrm{D}_{5/2}\ (\mathrm{F_e}=4)\}/2 = 385\ 284\ 566\ 370.4\ \mathrm{kHz}$$

$u_c/y = 5.2 \times 10^{-12}$.

From [5] cited in Table 4,

$$f\{^{87}\mathrm{Rb},\ 5\mathrm{S}_{1/2}\ (\mathrm{F_g}=2) - 5\mathrm{D}_{5/2}\ (\mathrm{F_e}=4)\}/2 - f\{^{85}\mathrm{Rb},\ 5\mathrm{S}_{1/2}\ (\mathrm{F_g}=3) - 5\mathrm{D}_{5/2}\ (\mathrm{F_e}=5)\}/2 = -576\ 002\ \mathrm{kHz}$$

$u_c/y = 1.3 \times 10^{-11}$,

one obtains

$$f\{^{85}\mathrm{Rb},\ 5\mathrm{S}_{1/2}\ (\mathrm{F_g}=3) - 5\mathrm{D}_{5/2}\ (\mathrm{F_e}=5)\}/2 = 385\ 285\ 142\ 371.40$$

$u_c/y = 1.4 \times 10^{-11}$.

3. Absolute frequency of the other transitions related to those adopted as recommended and frequency intervals between transitions and hyperfine components

These tables replace those published in BIPM Com. Cons. Long., 2001, **10**, 184-187 and *Metrologia*, 2003, **40**, 127-128.

The notation for the transitions and the components is that used in the source references. The values adopted for the frequency intervals are the weighted means of the values given in the references.

For the uncertainties, account has been taken of:

- the uncertainties given by the authors;
- the spread in the different determinations of a single component;
- the effect of any perturbing components;
- the difference between the calculated and the measured values.

In the tables, u_c represents the estimated combined standard uncertainty (1σ).

When a two-photon transition is listed, the listed frequency indicates the one-photon laser frequency.

Table 1

$\lambda \approx 778$ nm ^{85}Rb $5S_{1/2} - 5D_{3/2}$ two-photon transition		
$F_g - F_e$	$[f(5S_{1/2}(F_g) - 5D_{5/2}(F_e))/2 - f_{ref}$ / kHz]	u_c/ kHz
3−1	−44 462 655	7
3−2	−44 459 151	7
3−3	−44 453 175	7
3−4	−44 443 871	7
2−1	−42 944 789	7
2−2	−42 941 283	7
2−3	−42 935 308	7
2−4	−42 926 004	7
Frequency referenced to	$(5S_{1/2}(F_g = 3) - 5D_{5/2}(F_e = 5))/2\{^{85}Rb\}$: $f_{ref} = 385\,285\,142\,375$ kHz [6]	

Ref. [7]

Table 2

$\lambda \approx 778$ nm ^{85}Rb $5S_{1/2} - 5D_{5/2}$ two-photon transition		
$F_g - F_e$	$[f(5S_{1/2}(F_g) - 5D_{5/2}(F_e))/2 - f_{ref}$/ kHz]	u_c/ kHz
3–5	0	—
3–4	4718	9
3–3	9228	9
3–2	13 031	9
3–1	15 771	14
2–4	1 522 595	9
2–3	1 527 094	9
2–2	1 530 887	9
2–1	1 533 631	11
2–0	1 535 084	26
Frequency referenced to	$(5S_{1/2}(F_g = 3) - 5D_{5/2}(F_e = 5))/2\{^{85}Rb\}$: f(ref) = 385 285 142 375 kHz [6]	

Ref. [5][(1)], [7]

[(1)] Improved interval measurements are available for certain components and can be used provided appropriate consideration to uncertainties is made.

Table 3

$\lambda \approx 778$ nm ^{85}Rb $5S_{1/2} - 5D_{3/2}$ two-photon transition		
$F_g - F_e$	$[f(5S_{1/2}(F_g) - 5D_{5/2}(F_e))/2 - f_{ref}$/ kHz]	u_c/ kHz
2–0	–45 047 389	7
2–1	–45 040 639	7
2–2	–45 026 674	7
2–3	–45 004 563	7
1–1	–41 623 297	7
1–2	–41 609 335	7
1–3	–41 587 223	7
Frequency referenced to	$(5S_{1/2}(F_g = 3) - 5D_{5/2}(F_e = 5))/2\{^{85}Rb\}$: f(ref) = 385 285 142 375 kHz [6]	

Ref. [7]

Table 4

$\lambda \approx 778$ nm ^{85}Rb $5S_{1/2} - 5D_{5/2}$ two-photon transition		
$F_g - F_e$	$[f(5S_{1/2}(F_g) - 5D_{5/2}(F_e))/2 - f_{ref}$/ kHz]	u_c/ kHz
2–4	–576 001	9
2–3	–561 589	9
2–2	–550 112	9
2–1	–542 142	9
1–3	2 855 755	9
1–2	2 867 233	9

1−1	2 875 200	9

Frequency referenced to $\left(5S_{1/2}\left(F_g=3\right)-5D_{5/2}\left(F_e=5\right)\right)/2\left\{^{85}Rb\right\}$: $f\,(\mathrm{ref}) = 385\ 285\ 142\ 375$ kHz [6]

Ref. [5][(1)], [7]

4. Absolute frequency of other transitions

4.1. Absorbing atom ^{87}Rb, $5S_{1/2}\ (F_g = 2) - 7S_{1/2}\ (F_e = 2)$ two-photon transition

The values $f = 394\ 397\ 384\ 460$ kHz
$\lambda = 760\ 127\ 906.05$ fm

with a relative standard uncertainty of 1.7×10^{-10} apply to the single-photon laser frequency of the two-photon transition.

Adopted value $f = 394\ 397\ 384\ 460\ (67)$ kHz
$u_c/y = 1.7 \times 10^{-10}$

for which:
$\lambda = 760\ 127\ 906.05\ (.13)$ fm
$u_c/y = 1.7 \times 10^{-10}$

After [Refs [8], [9]]

4.2. Absorbing atom ^{87}Rb, $5S_{1/2}\ (F_g = 1) - 7S_{1/2}\ (F_e = 1)$ two-photon transition

The values $f = 394\ 400\ 482\ 100$ kHz
$\lambda = 760\ 121\ 936.0$ fm

with a relative standard uncertainty of 4.5×10^{-10} apply to the single-photon laser frequency of the two-photon transition.

Adopted value $f = 394\ 400\ 482\ 100\ (180)$ kHz
$u_c/y = 4.5 \times 10^{-10}$

for which:
$\lambda = 760\ 121\ 936.0\ (.34)$ fm
$u_c/y = 4.5 \times 10^{-10}$

After [Refs [8], [9]]

Absorbing atom ^{85}Rb, $5S_{1/2}\left(F_g=3\right)-5D_{5/2}\left(F_e=5\right)$ **two-photon transition**

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